Verification and Evaluation Plan

Purpose and scope

This plan explains how achievement of the MathVerity aim will be assessed: helping students in grades 5–6 develop selected mathematical skills and giving parents understandable evidence of progress. It covers the planned eight skills, 120 original items and Kazakh and Russian interfaces. All measures and targets below are proposed; completed results will be reported separately.

Verification asks whether the platform behaves as required. Evaluation asks whether its outcomes are useful to learners and parents. Both are needed: correct operation alone does not demonstrate improved learning.

Verification scenarios

Review each mandatory requirement through a user scenario. Record the action, expected behaviour, observed result and supporting evidence. The following scenarios cover the main project promises.

Area	What will be verified	Evidence required
Mathematical feedback	Supported equivalent numeric and fractional answers receive the correct result.	Reference answer and recorded feedback agree.
Practice and assessment	Practice offers up to three hints. Independent work offers none and retains the first valid answer.	Walkthrough showing both modes and their recorded answers.
Progress and rewards	Assisted answers and XP do not confirm a skill. Statuses reflect the documented evidence rule.	Examples of insufficient, developing, confirmed and old evidence.
Parent access	Parents see only a child linked to their own family. Revocation removes access.	Linked, unlinked and revoked-access scenarios.
Bilingual content	Both languages provide equivalent tasks, explanations and hints. The other developer reviews each version.	Paired language versions and a recorded peer review.

Evaluation questions and outcome measures

Evaluation focuses on independent mathematical performance, the meaning of skill statuses and the usefulness of parent reports. Activity and rewards provide supporting context. They cannot substitute for evidence that a learner can solve a new item without built-in assistance.

Question	Measure and interpretation
Does independent performance improve?	Calculate independent accuracy as correct first answers divided by all assigned assessment items, multiplied by 100. Report unanswered items as incorrect when the assessment is attended. Compare scores before and after practice using different, comparable items.
Can the learner apply the skill?	Measure accuracy on new items that change the context or structure. Include a structured explanation of a solution step. Success on a repeated practice item is weaker evidence of transfer.
Do statuses match the evidence?	Both developers independently review the recorded answers against the fixed status rule. Compare the system status with the resolved review label. The provisional target is at least 90% agreement; report disagreements.
Can parents understand the report?	Ask parents to identify a skill needing review, explain the supporting evidence and choose a next step. Score first responses against a prepared answer key. Target at least 80% correct responses and 80% of parents identifying both the skill and next step.
Are hints suitable?	Both developers review hints for mathematical correctness, clarity, requested language and absence of the final answer. Target at least 95% fully acceptable hints in each language, with no unresolved disclosure of answers or personal data.

Interpreting learning outcomes

For each learner, learning gain is the later independent score minus the earlier score, expressed in percentage points. Report actual scores and the number of learners with usable paired results. A later independent check can indicate whether performance is retained. Report missing assessments separately instead of treating an absent learner as having scored zero.

Baseline for comparison

The baseline is MathVerity with the team's reviewed static hints. The comparison version uses restricted AI hints during practice and returns to the same static hints when AI is unavailable. Independent assessment provides no hints in either version.

Keep the interfaces, languages, item difficulty, scoring, skill-status rules, reports and gamification the same. Use comparable practice exposure and assessment conditions. This makes the practice hint source the intended difference between the versions.

Compare the change in independent scores for the AI-hint version with the change for the static-hint version. A difference of at least five percentage points is a provisional practical signal. Report the difference and its uncertainty. An uncertain or negative result must be recorded; higher activity or XP cannot establish an AI learning benefit.

Measurement conditions

Use the same scoring rules and comparable item difficulty for every measurement. Earlier and later assessments must use different items assessing the same skills. Keep the interface language consistent for each learner and record the device, setting and any interruption or outside assistance.

Independent assessment means no built-in hints. Where outside help cannot be ruled out, state that limitation. Record hint use, static fallback and completed items so that differences in support or practice exposure remain visible.

Freeze the assessed content, scoring criteria and status rule before collecting results. Report Kazakh and Russian results separately as well as overall. Use participant codes in evaluation records and report aggregate findings. Participation involving children requires parental consent and learner assent.

How achievement of the aim will be demonstrated

The final conclusion will connect each part of the project aim to observable evidence. Requirements verification demonstrates that the intended workflows exist and behave correctly. Evaluation demonstrates what learners and parents can achieve with those workflows.

Part of the aim	Evidence to present	Conclusion supported
Provide a usable learning platform	Completed verification scenarios, both interface languages and reviewed original content.	The demonstrated workflows meet the stated requirements.
Improve mathematical skills	Independent scores before and after practice, transfer performance and comparison with static hints.	Observed improvement and the strength or limits of evidence for added AI benefit.
Explain progress to parents	Report-task answers, evidence interpretations and next-step choices.	Whether the reviewed reports are understandable and actionable.
Give evidence-based statuses	Recorded answer evidence, manual review labels and status disagreements.	Consistency with the documented rule, rather than external proof of mastery.

Completion criteria and records

All mandatory requirements must pass verification. Unresolved incorrect scoring, misleading confirmation or unauthorised family access prevents a positive verification conclusion. Record each requirement as passed, failed, blocked or not yet reviewed, with a link to its evidence.

The evaluation report will contain the measurements, baseline comparison, actual denominators, limitations and proposed improvements. Meeting a provisional target supports only the corresponding claim. A functioning platform and an honest evaluation may complete the project even when an additional benefit from AI is not demonstrated.